

Benchmarking LLMs via Uncertainty Quantification

Fanghua Ye Mingming Yang Jianhui Pang Longyue Wang
Derek F. Wong Emine Yilmaz Shuming Shi Zhaopeng Tu

Tencent AI Lab · UCL · University of Macau

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Objective

Problem

Current LLM leaderboards (HuggingFace, OpenCompass . . .) rank models by **accuracy only**.

Two models with *equal accuracy* may differ greatly in **uncertainty** – yet benchmarks treat them identically.

Goal

Integrate **uncertainty quantification** into LLM benchmarking via **conformal prediction** – a statistically rigorous, distribution-free, model-agnostic method.

Objective

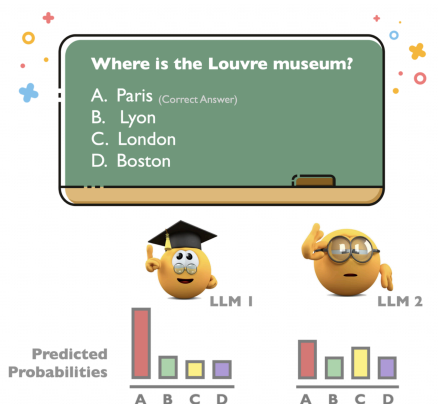


Figure: An illustration of two LLMs accurately predicting the true answer (with option A possessing the highest probability, but showing different levels of uncertainty. Note that when both LLMs predict a wrong answer, they may also display different levels of uncertainty.

Tasks & Datasets

5 NLP Tasks – all reformulated as 6-option MCQA (A–F)

Task	Dataset	Input	Example question
QA	MMLU	Question only	<i>“What is the powerhouse of the cell?”</i>
RC	CosmosQA	Context + Q	<i>“Based on the passage, why did she leave?”</i>
CI	HellaSwag	Context + Q	<i>“A man starts boiling water. . . what happens next?”</i>
DRS	HaluEval-Dial	Dialogue + Q	<i>“Which response best continues this conversation?”</i>
DS	HaluEval-Sum	Document + Q	<i>“Which summary is accurate and non-hallucinated?”</i>

6 options per question

A, B, C, D = content options **E** = *“I don’t know”* **F** = *“None of the above”*

Each dataset has **10,000 instances**; split 50% cal / 50% test.

Prompting Strategies

Base Prompt – question + choices, no instruction

Question: What is the capital of France?

Choices: A. Berlin B. Paris C. Madrid D. Rome E. I don't know F. None of the above

Answer:

Shared Instruction Prompt – general task description prepended

The following is a multiple-choice question with six potential answers. Only one is correct. You only need to output the option.

Question: What is the capital of France? ... Answer:

Task-specific Instruction Prompt – task-tailored description

The following is a multiple-choice question about question answering. Answer based on your world knowledge. You only need to output the option.

Question: What is the capital of France? ... Answer:

Results averaged over all 3 strategies to reduce prompt sensitivity.

How Logits Are Obtained from LLMs

Open-source (HuggingFace)

- 1 Run one **forward pass** on the prompt
 - 2 Extract logits at the **last token position**
 - 3 Slice values for tokens A, B, C, D, E, F
 - 4 Apply **softmax** $\rightarrow P(X)$
- $\Rightarrow$ 1 call, exact vocabulary distribution

Closed-source (GPT-3.5/4)

- 1 **Sample 50 answers** per question
 - 2 Count frequency of each option
 - 3 Normalize $\rightarrow$ estimated $P(X)$
 - 4 Apply temperature scaling to avoid zero probs
- $\Rightarrow$ 50 calls, approximation
(JSD = 0.05)

Our implementation (Ollama / qwen3.5:4b)

Ollama does not expose full vocabulary logits $\Rightarrow$ **6 API calls** per question:
append " A", " B", ..., " F" to prompt $\rightarrow$ read logprob of that continuation token

Methodology – Conformal Prediction

Coverage Guarantee

For test instance X_t with true label Y_t , CP outputs a **prediction set** $\mathcal{C}(X_t) \subseteq \mathcal{Y}$ such that:

$$P(Y_t \in \mathcal{C}(X_t)) \geq 1 - \alpha$$

α = user-specified error rate (e.g. $\alpha = 0.1 \Rightarrow$ **90 %** coverage guarantee).

Uncertainty Measure

Set Size (SS) = average $|\mathcal{C}(X_t)|$ over all test instances.

Small SS $\Rightarrow$ **confident model** **Large SS** $\Rightarrow$ **uncertain model**

Methodology – Score Functions

LAC – Least Ambiguous Classifier

$$s(X, Y) = 1 - f(X)_Y$$

Uses softmax score of the true label only.

⇒ tends to smaller sets.

APS – Adaptive Prediction Sets

$$s(X, Y) = \sum_{\substack{Y' \in \mathcal{Y}: \\ f(X)_{Y'} \geq f(X)_Y}} f(X)_{Y'}$$

Accumulates ranked softmax scores.

⇒ better on hard instances.

*Final metrics averaged over **LAC** and **APS**.*

Algorithm – CP for LLMs

Algorithm 1: Conformal Prediction for LLMs Input: $\mathcal{M}, \mathcal{D}, \alpha$ Output: Acc, SS, CR

```
1 for each prompt strategy  $p \in \{\text{base, shared, task}\}$  do // 3 strategies
2   for each ICL setting  $k \in \{0\text{-shot, few-shot}\}$  do // 2 ICL modes
3      $P(X) \leftarrow \text{Softmax}(\text{logits}_{\mathcal{M}}(X_{p,k}))$  for all  $X \in \mathcal{D}$  // query LLM
4      $\mathcal{D} \rightarrow \mathcal{D}_{\text{cal}} (50\%) + \mathcal{D}_{\text{test}} (50\%)$  // random split
5     for each  $s \in \{\text{LAC, APS}\}$  do // 2 score fns
6        $s_i \leftarrow s(X_c^{(i)}, Y_c^{(i)})$  for all  $i \in \mathcal{D}_{\text{cal}}$  // nonconformity scores
7        $\hat{q} \leftarrow \text{Quantile}\left(\{s_i\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$  // adaptive threshold
8        $\mathcal{C}(X_t) \leftarrow \{Y' \in \mathcal{Y} : s(X_t, Y') \leq \hat{q}\}$  for all  $X_t \in \mathcal{D}_{\text{test}}$  // prediction sets
9       if  $\mathcal{C}(X_t) = \emptyset$  then keep  $\arg \max_{Y'} P(X_t)_{Y'}$ 
10 return avg over  $p, k, s$ : Acc, SS, CR
```

Methodology – Evaluation Metrics

Metric	Formula	Meaning
Acc	$\frac{1}{ \mathcal{D}_{\text{test}} } \sum \mathbf{1}[Y_p = Y_t]$	Prediction accuracy
SS	$\frac{1}{ \mathcal{D}_{\text{test}} } \sum \mathcal{C}(X_t) $	Uncertainty ($\uparrow$ more uncertain)
CR	$\frac{1}{ \mathcal{D}_{\text{test}} } \sum \mathbf{1}[Y_t \in \mathcal{C}(X_t)]$	Coverage ($\geq 1-\alpha$ guaranteed)

Results – Coverage Rate CR (%) per Task

Source: Table 1 – mean of LAC & APS Target: CR $\geq$ 90%

LLM	QA	RC	CI	DRS	DS	Avg
Qwen-14B	92.58	95.20	95.29	91.92	89.71	92.94
Yi-6B	91.30	94.06	92.35	91.36	91.38	92.09
Gemma-7B	93.57	94.16	92.13	91.59	90.70	92.43
Mistral-7B	93.00	92.91	89.94	91.02	92.05	91.78
Llama-2-13B	92.59	93.15	90.50	90.92	90.82	91.60
Qwen-7B	92.79	94.02	91.53	92.43	89.56	92.07
InternLM-7B	90.68	93.28	90.10	90.40	90.34	90.96
Llama-2-7B	91.37	90.69	90.97	89.60	90.04	90.53
DeepSeek-7B	91.18	89.95	90.16	90.89	90.21	90.48
MPT-7B	89.79	90.54	90.12	90.80	89.71	90.19
Falcon-7B	90.04	89.95	89.82	90.46	90.71	90.19

Results – Prediction Accuracy Acc (%) per Task

Source: Table 1 – mean of LAC & APS

LLM	QA	RC	CI	DRS	DS	Avg
Qwen-14B	64.25	91.52	91.00	73.90	49.33	74.00
Yi-6B	57.57	85.99	76.50	58.72	66.06	68.97
Gemma-7B	62.24	85.29	73.58	66.79	40.80	65.74
Mistral-7B	60.44	81.94	62.93	53.21	62.16	64.14
Llama-2-13B	52.52	77.23	59.66	52.65	60.05	60.42
Qwen-7B	55.21	83.89	63.70	64.04	32.53	59.87
InternLM-7B	48.37	73.86	46.21	43.72	34.38	49.31
Llama-2-7B	45.60	65.79	43.05	32.61	45.60	46.53
DeepSeek-7B	45.65	65.39	42.66	33.50	42.15	45.87
MPT-7B	29.49	31.69	25.50	24.38	24.86	27.18
Falcon-7B	23.75	24.98	24.91	25.86	24.69	24.84

Results – Prediction Uncertainty SS per Task

Source: Table 1 – mean of LAC & APS Lower SS = more certain

LLM	QA	RC	CI	DRS	DS	Avg
Qwen-14B	2.80	1.74	2.02	1.94	2.37	2.17
Yi-6B	3.20	1.92	1.88	2.85	1.96	2.36
Gemma-7B	2.72	1.88	2.04	2.14	3.11	2.38
Mistral-7B	2.80	1.75	2.48	2.71	2.40	2.43
Llama-2-13B	3.06	2.24	2.72	2.55	2.24	2.56
Qwen-7B	3.26	2.15	2.28	2.51	2.92	2.63
InternLM-7B	3.49	2.19	3.28	3.63	4.47	3.41
Llama-2-7B	3.20	2.39	3.27	3.26	3.30	3.09
DeepSeek-7B	3.34	2.77	3.06	3.40	3.08	3.13
MPT-7B	3.53	3.46	3.60	3.59	3.66	3.57
Falcon-7B	3.90	3.60	3.66	3.64	3.92	3.75

Results – Three Key Findings

Finding 1: Higher Accuracy $\neq$ Lower Uncertainty

InternLM-7B outperforms MPT-7B by **+19.34 %** Acc on DRS, yet shows **higher** SS (3.63 vs. 3.59). Rankings by Acc and SS diverge across *all* tasks.

Finding 2: Larger $\neq$ More Certain

Qwen 1.8B $\rightarrow$ 14B: Acc and certainty both improve.

Qwen 14B $\rightarrow$ 72B: Acc rises, but SS **increases** on RC & DRS – non-monotonic.

Finding 3: Instruction Finetuning Raises Uncertainty

Chat-V1 and Chat-V2 show **consistently higher SS** than base models across Llama-2, Yi, DeepSeek, and Falcon series.

Results – Closed-Source LLMs (Table 4)

MMLU (QA task) only – sampling-based probability estimation

LLM	CR (%)	Acc (%)↑	SS↓
GPT-4	90.41	81.75	1.65
GPT-3.5	89.98	62.99	3.05

Key observations

- **GPT-4** achieves highest Acc (81.75%) and lowest SS (1.65) – most accurate and certain.
- **GPT-3.5** CR falls below 90% – sampling approximation less reliable.

Results – CP vs. Perplexity (InternLM-7B)

Table 2: Coverage Rate (%) & Set Size – target CR $\geq$ 90 %

Task	CR (%)		SS	
	CP	PPL	CP	PPL
QA	90.68	83.44	3.49	2.89
RC	93.28	95.48	2.19	2.39
CI	90.10	96.25	3.28	3.97
DRS	90.40	86.80	3.63	3.42
DS	90.34	87.13	4.47	4.33
Avg	90.96	89.82	3.41	3.40

PPL **fails** 90 % on QA, DRS, DS. CP **always meets** the coverage guarantee.

Results – ECE Comparison (InternLM-7B)

Table 3: Expected Calibration Error (%) – lower is better

Task	CP	Entropy	$P_{\max}$
QA	15.83	15.83	15.83
RC	1.33	1.32	1.41
CI	3.16	3.45	3.75
DRS	12.11	12.40	12.45
DS	9.30	9.30	9.62
Avg	8.35	8.46	8.61

CP achieves the **lowest average ECE** across all methods.

Our Experiment – Qwen3.5-4B on MMLU

Setup

- Model: qwen3.5:4b via Ollama (local inference)
- Dataset: MMLU ($n = 200$ test samples, 1-shot, base prompt)
- Logits via continuation logprob scoring (6 API calls/question)
- CP parameters: $\alpha = 0.1$, cal/test split 50 %

Results

Model	Acc (%) $\uparrow$	SS $\downarrow$	CR (%) ≥ 90
Qwen3.5-4B	53.40	3.96	93.69

CR ≥ 90 % target **met.** SS = 3.96 reflects moderate uncertainty on MMLU.

Conclusion & Future Work

Conclusion

- Accuracy alone is **insufficient** for LLM evaluation.
- Conformal prediction offers a **rigorous, model-agnostic** uncertainty measure.
- Three findings: $\text{Acc} \neq \text{certainty}$; larger $\neq$ more certain; finetuning raises uncertainty.

Future Work

- Extend CP to **free-form text generation** beyond MCQA.
- Apply to **multi-modal** and **closed-source** LLMs (GPT-4 etc.).
- Integrate uncertainty into **RAG** and **in-context learning** pipelines.